GROUP WORK PROJECT # 1
GROUP NUMBER: 7381

MScFE 622: Stochastic Modeling

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
Team member 1	Mehedi Hasan
Team member 2	
Team member 3	

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. Note: You may be required to provide proof of your outreach to non-contributing members upon request.
We tried to reach him several times for him to take part of the group work but he ignored all calls or text message.

STEP 1

Question 1

Q1.1 Calibration Process for classic Heston (1993) model (without jumps)

Step 1: Data Selection and Preprocessing

- Uploaded market option data and selected options with exactly 15 days to maturity, aligning with the client's preference for short-term securities.
- Set the risk-free rate to 1.50% for valuation.

Step 2: Defining the Heston Model Characteristic Function

- Implemented the Heston model's characteristic function using Lewis' (2001) Fourier-based approach.
- Applied the Fourier integral transformation to derive the theoretical call option value.\

Step 3: Model Calibration and Error Minimization

- Used Mean Squared Error (MSE) to measure the difference between observed market prices and model prices.
- Ran an optimization process (brute force) to minimize MSE and calibrate Heston model parameters.
- **Calibrated Parameters**
  - kappa_v = 3.0473
  - theta_v = 0.1245
  - sigma_v = 0.000004109
  - rho = 0.6156
  - Vo = 0.1034

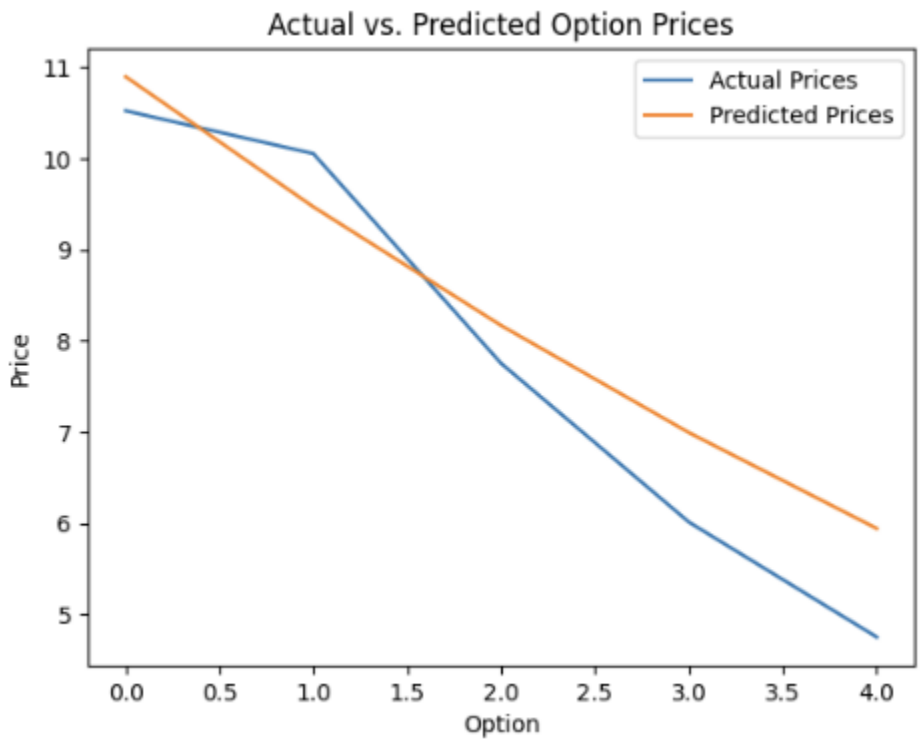
Step 4: Pricing the Asian Option

- Selected an Out-of-the-Money (OTM) Asian option with strike price K=237.5
- Used Monte Carlo simulations to determine the Asian call option price:
  - Asian Call Option Price = $2.58.

Step 5: Pricing the Asian Put Option Using Put-Call Parity

- Applied Put-Call Parity to compute the put option price.
- Asian Put Option Price = $7.17

Graph of the predicted call prices against the actual call prices



The graph compares Actual vs. Predicted Option Prices, illustrating discrepancies between market-observed prices (blue line) and model-predicted prices (orange line). The actual prices decrease more sharply, indicating that the model underestimates price drops for deeper out-of-the-money options. This suggests a calibration issue or the need for a refined model incorporating additional market dynamics.

Q 1.3: ATM Asian Call Option Pricing Using Monte Carlo Simulation

Step 1: Defining the Option Parameters

- Initial Stock Price (S_0) = 232.90
- Strike Price (K) = S_0 (ATM option)

- Maturity (T) = 20 days (converted to years)
- Risk-Free Rate (r) = 1.15%
- Time Steps (n_timesteps) = 20
- Simulations (n_simulations) = 100,000

Step 2: Monte Carlo Simulation using Heston Model

- Discretized the Heston model to simulate stock price and variance paths.
- Iterated over 100,000 simulations with 20 time steps each.
- Computed the average stock price over the time horizon to determine the Asian option payoff.

Step 3: Calculating the Fair Price

- Discounted the expected Asian call option payoff using the risk-free rate.
- Obtained the fair price of the ATM Asian call option.

Step 4: Adding a 4% Fee for Final Pricing

- Applied a 4% markup to account for the bank's profit margin:
  Final Price=Fair Price×(1+0.04)
- This final value represents the client's purchase price.

Fair price of the Asian call option: 2.58

Final price for the client (including 4% fee): 2.68

Fair price of the Asian call option: 2.36

Final price for the client (including 4% fee): 2.45

Both the prices are close.

Step 2

Q2.1 for the case at hand (60-day maturity instrument) using a Heston model with jumps (i.e., Bates, 1996 model).with Lewis(2001) approach

Step 1: Bates Model Formulation

- The Bates model extends the Heston model by incorporating jumps in asset prices.
- It combines stochastic volatility (Heston) with jump diffusion (Merton).

Step 2: Bates Model Characteristic Function

- The characteristic function accounts for both volatility and jump components.
- Used Lewis (2001) inverse Fourier transform approach to price options.
- The characteristic function $\phi(u)$ is derived by multiplying:
    - Heston model's characteristic function ($\phi H(u)$)
    - Jump diffusion component from Merton model

Step 3: Asian Option Pricing Characteristic Function

- Instead of using the final stock price STS_TST, the characteristic function is adjusted for the average price of the underlying asset.
- This allows direct computation of Asian option values under the Bates model.

Step 4: Calibration of the Bates Model

- **Step 1: Initial Guess**
    - Used initial estimates for Heston parameters: kappa_v, theta_v, sigma_v, rho
    - Additional jump parameters estimated: lambda, mu_j, sigma_j
- **Step 2: Error Function Definition**
    - Defined Mean Squared Error (MSE) to measure the difference between market prices and model-generated prices.

- **Step 3: Parameter Optimization**
  - Applied an optimization algorithm to minimize MSE and calibrate the model.
  - Retained previously imported option data and adjusted Bates parameters accordingly.

Calibrated Bates parameters:

Using κ, θ, σ_v, ρ, v_0, λ Heston-Lewis parameters:

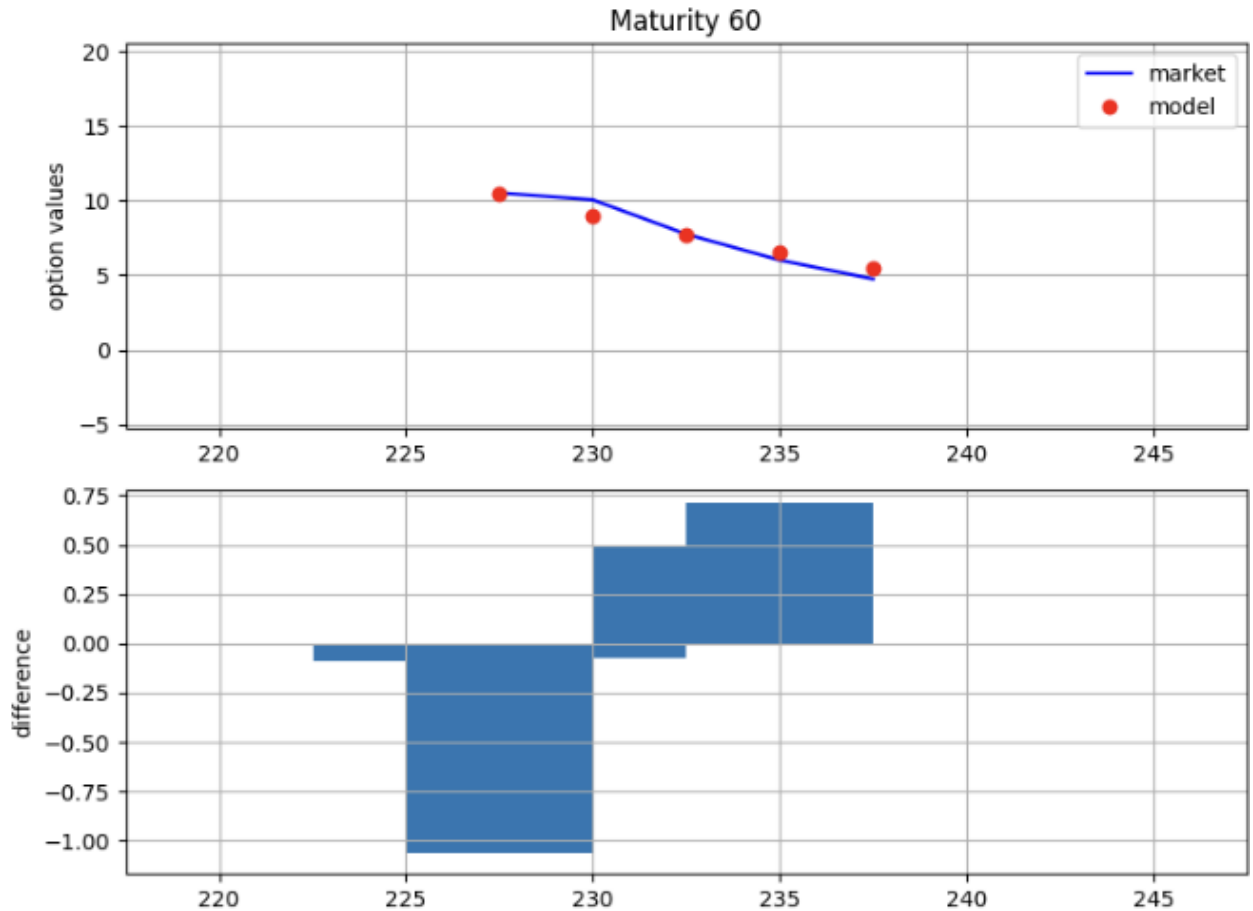
```
p0 = [kappa_v, theta_v, sigma_v, rho, v0, lamb, mu, delta]
```

= (3.047, 0.125, 4.108e-06, 0.615, 0.1033, 0.0, -0.04, 3.88)

Using κ, θ, σ_v, ρ, v_0, λ Heston-Carr Madan parameters

```
kappa_v, theta_v, sigma_v, rho, v0, lamb_jump, mu_jump, delta_jump
```

= (4.226, 0.1277, 9.1344e-07, 0.4588, 0.1055, 0.0, -0.02, 3.97)



The top chart compares market option values (blue line) with model-predicted values (red dots) for a 60-day maturity instrument. The model closely follows the market prices but shows slight deviations at certain points.

The bottom chart visualizes the difference between market and model prices, highlighting overestimations and underestimations. Negative bars indicate the model underpricing options, while positive bars indicate overpricing. This suggests minor calibration adjustments may be needed for improved accuracy.

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THE Call option value using Bates-heston-lewis parameters: 5.37
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Q 2.2 The case at hand (60-day maturity instrument) using a Heston model with jumps (i.e., Bates, 1996 model). with Carr-Madan approach

Step 1: Bates Model Overview

- The Bates model extends the Heston stochastic volatility model by incorporating a Merton-type jump-diffusion process.
- This combination accounts for both random fluctuations in volatility and sudden jumps in asset prices.

Step 2: Fourier Transform in the Carr-Madan Approach

- The Carr-Madan method computes option prices using the Fourier transform of a damped payoff function to ensure convergence.
- Fast Fourier Transform (FFT) is applied to efficiently compute the call option price integral as per Carr & Madan (1999).

Step 3: Calibration of the Bates Model

- Step 1: Initial Guess
    - Set initial parameter estimates for the Bates model.
- Step 2: Error Function Definition
    - Defined Mean Squared Error (MSE) to quantify the difference between observed market prices and model prices.
- Step 3: Parameter Optimization
    - Used an optimization algorithm to minimize MSE and refine parameter estimates.

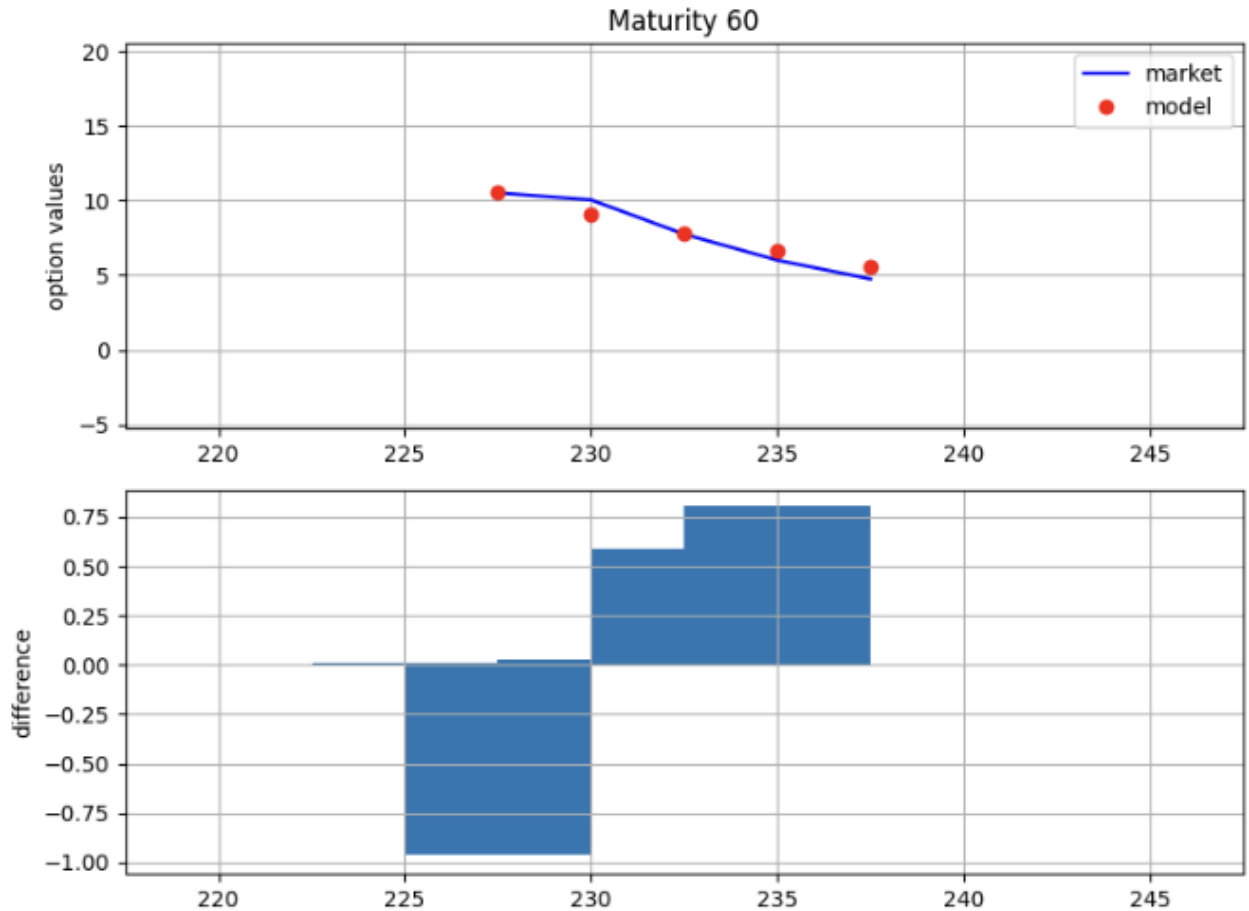
Calibrated Bates parameters:

Using κ, θ, σ_v, ρ, v_0, λ Heston -Lewis parameters:

$[\kappa, \theta, \sigma_v, \rho, v_0, \lambda, \mu_J, \sigma_J]$ = [3.047, 0.125, 4.108e-06, 0.615, 0.1033, 0.02121, -0.5999, 4.293e-07]

Using κ, θ, σ_v, ρ, v_0, λ Heston-Carr Madan parameters:

[κ, θ, σ_v, ρ, v_0, λ, μ_J, σ_J] = [4.226, 0.1277, 9.1344e-07, 0.4588, 0.1055, 0.0, 0.0, 0.0]

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THe CAll option price using Bates-Heston-Carr Madan approach is  5.4781
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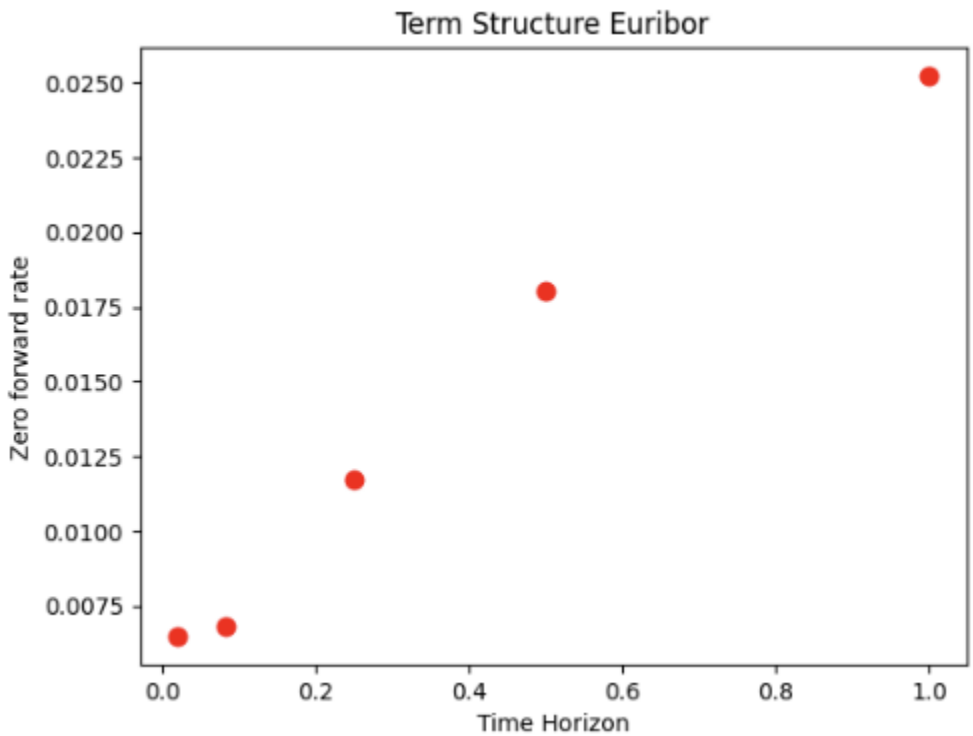
STEP 3

a)Calibration of CIR(1985) model using Euribor for different maturities

Step 1: Data Selection and Processing

- Euribor interest rates and their corresponding maturities were coded into a NumPy array.
- Defined current interest rates, capitalization factors, and zero-forward rates based on the observed Euribor data.

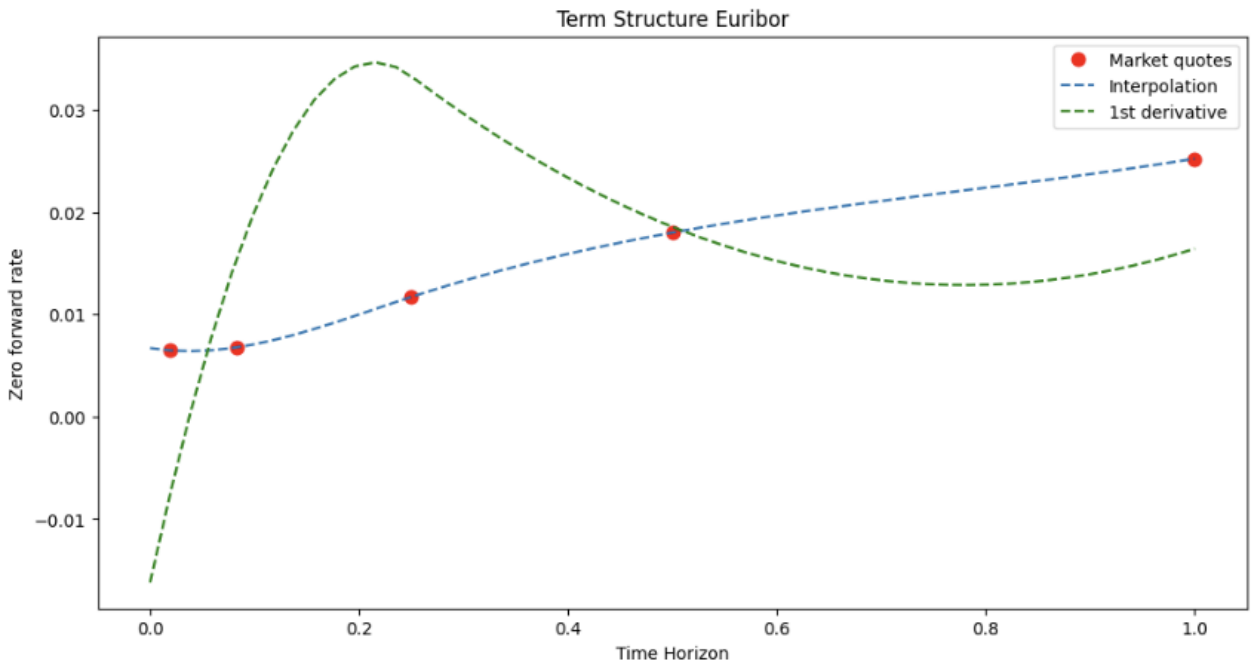
- Plotted forward rates vs. maturities, revealing a linear positive relationship in line with the term structure of interest rates (Matacz & Bouchaud, 2000), where longer maturities correspond to higher forward rates.



The graph represents the term structure of Euribor, showing the zero forward rate against the time horizon. The positive trend indicates that longer maturities correspond to higher forward rates, aligning with the theory of the term structure of interest rates. This suggests an upward-sloping yield curve, reflecting market expectations of rising interest rates over time.

Step 2: Interpolation of Market Rates

- The dataset contained only 5 data points, which was insufficient for accurate calibration.
- To enhance data granularity, we applied cubic spline interpolation to estimate weekly rates over a 1-year period, increasing the data points to 52.
- A comparison graph of interpolated vs. market data showed minimal deviation, confirming the reliability of the interpolation.



The graph visualizes the term structure of Euribor, comparing market quotes (red dots) with interpolated rates (blue dashed line) obtained through cubic spline interpolation. The first derivative (green dashed line) represents the rate of change in forward rates. The interpolation closely follows market data, ensuring a smooth estimation, while the derivative highlights variations in rate trends over time.

Step 3: Defining Forward Rates and Error Function in the CIR Model

- Defined the forward rates function based on the CIR (1985) model for parameter calibration.
- Although mathematically complex, Python was utilized to simplify computations and extract model parameters.
- Established the error function using Mean Squared Error (MSE) to measure the accuracy of predicted interest rates against observed market data.

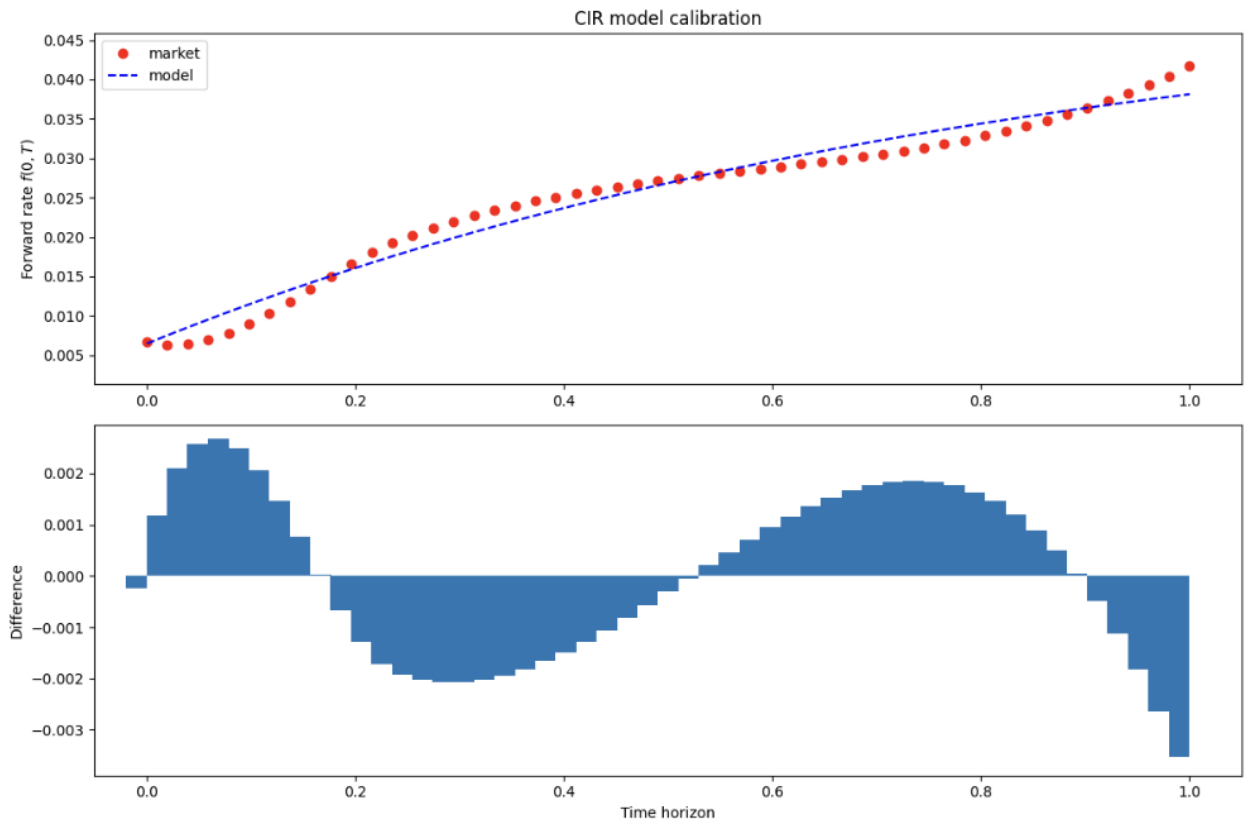
Step 4: Optimization and Calibration Results

- Implemented an optimization function to minimize the difference between model-predicted rates and market rates.
- Ran the calibration process to determine the CIR model parameters.

Calibration Results:

- Kr=0.999
- θr=0.107
- σr=0.001

Graphs of predicted rates from our CIR model and Market rates



The top chart compares market forward rates (red dots) with CIR model-predicted rates (blue dashed line) after calibration. The model closely follows market rates but shows slight deviations at different maturities.

The bottom chart illustrates the difference between market and model rates, highlighting periods of overestimation and underestimation. The oscillating pattern suggests potential refinements in calibration to improve accuracy.

b) Forecasting Euribor Rates Using the CIR Model

- 95% Confidence Interval: Euribor rates over the next 12 months are expected to range between 0.0700 and 0.0706.
- Expected Value: The predicted Euribor rate for the next year is 0.0703.
- Pricing Implication:
  - Using the current interest rate (0.02556) instead of the model-predicted rate (0.0703) will overprice the product.
  - The CIR model suggests that the predicted rates would result in underpricing (Grzelak & Oosterlee, 2011).

References

Matacz, A., & Bouchaud, J. P. (2000). An empirical investigation of the forward interest rate term structure. *International Journal of Theoretical and Applied Finance*, *3*(04), 703-729.

Escobar, M., & Gschnaidtner, C. (2016). Parameters recovery via calibration in the Heston model: A comprehensive review. *Wilmott*, *2016*(86), 60-81.

Grzelak, L. A., & Oosterlee, C. W. (2011). On the Heston model with stochastic interest rates. *SIAM Journal on Financial Mathematics*, *2*(1), 255-286.

Beyazit, M. F., & Eroglu, K. I. (2021). Lewis Model Revisited: Option Pricing with Lévy Processes. *Bulletin of the Malaysian Mathematical Sciences Society*, *44*, 1653-1668.